Fiscal Year:	FY 2015	**Task Last Updated:**	FY 02/03/2015
PI Name:	Wyrobek, Andrew Ph.D.		
Project Title:	Molecular Characterization of Choroid Plexus and Hippocampal Damage and Degenerative CNS Risks from Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ajwyrobek@lbl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	925-989-4466
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	1 Cyclotron Rd MS: 74R0157		
PI Address 2:			
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-8099	**Congressional District:**	13
Comments:	For immediate assistance please contact my administrator Caron LaMarsh clamarsh@lbl.gov, 510.486.5317		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	04/04/2014	**End Date:**	04/03/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	6
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	February 2015: This is a renewal project and the key members are: Sandhya Bhatnagar, Life Sciences Division, LBNL; Halina E. Witkowska,University of California, San Francisco; Bernard Rabin, University of Maryland, Baltimore Campus ; Richard Britten, East Virginia Medical School; Tore Straume, NASA Ames, Space Biosciences Division; Leif E. Peterson, Houston Methodist Research Institute		
COI Name (Institution):	Peterson, Leif (The Methodist Hospital Research Institute) Straume, Tore (NASA Ames Research Center) Witkowska, Halina E. (University of California, San Francisco) Rabin, Bernard (University of Maryland, Baltimore Campus) Britten, Richard (Eastern Virginia Medical School) O'Banion, Kerry (University of Rochester)		
Grant/Contract No.:	NNJ14HP06I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A major goal of the NASA space radiation research program is to enable human exploration of space without exceeding limits for immediate and persistent risks to the central nervous system (CNS) from space radiation. Our research plan builds on the mounting evidence of the radiation-sensitivity of the hippocampus, which is important for memory and performance, and the choroid plexus (CP), which produces cerebral spinal fluid and maintains the blood-brain barrier. Using transcriptomic, proteomic, metabolomics, and immunohistochemical analyses of CNS tissue from rodents exposed to simulated space radiation, we have identified molecular and cellular changes associated with oxidative stress, neuro-inflammation, vasculature damage, changes in the protein composition of cerebral spinal fluid, and amyloid deposits such as those found in Alzheimer's disease (AD). We propose new research to investigate the low-fluence responses and post-exposure-time dependence of these findings, to identify the molecular mechanisms of inter-individual variations in behavioral responses after radiation exposure, and to validate the predicted mechanisms using tissue of irradiated and behaviorally characterized rodents. Our specific aims are to: (1) Investigate the cellular and functional abnormalities in the hippocampus and CP of HZE-irradiated rats that are predicted by our 'omics findings with the emphasis on defects associated with AD. (2) Investigate hippocampal and CP radiation damage responses associated with inter-individual variation in radiation-induced neurocognitive deficits, using N-glycoproteomic, metabolomics, and in situ analyses, with emphasis on tissue defects associated with AD. (3) Characterize the radiation sensitivity of persistent CP epithelial and endothelial damage in AD mouse models and examine their associations with neuro-inflammation and other defects associated with AD. This research is foundational to establish quantitative mechanism-based predictors for neuro-cognitive risks and onset of AD symptoms in individuals exposed to high-LET (linear energy transfer) space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is designed to provide NASA with a mechanistic understanding of the risks for impaired neurocognitive performance and individual susceptibility for neurological diseases after exposure to high energy heavy ions (HZE) radiation. Our approach: (a) employs state-of-the-art 'omics and tissue technologies to investigate CNS damage response functions in animals exposed to low-fluence HZE particles (i.e., <1 hit per cell), (b) links the type and magnitude of molecular defects with neurocognitive performance results, and (c) validates the predicted damage response functions in brain tissue. Our project focuses on radiation damage to the hippocampus (site of adult neurogenesis; spatial and performance memories) and choroid plexus (CP, small cauliflower-like tissue in the ventricles adjacent to the hippocampus that produces cerebral spinal fluid (CSF) and maintains blood-brain homeostasis). Our research strategy distinguishes between two broad categories of risk mechanisms: (1) mechanisms related to exposure dose, and (2) mechanisms of inter-individual variations in risks for neurocognitive performance and neurological diseases (i.e., Alzheimer's Disease (AD). Our research is conducted in outbred strains of rats and in double and triple mutation AD mice. Successful completion of this project will provide a mechanistic foundation for space-radiation-induced CNS risk that is based on quantitative molecular changes and their associations with measured neurocognitive deficits. The project considers short term and persistent effects, as well as inter-individual differences in responses. The project will employ space-relevant fluences of simulated HZE radiation, ensuring relevance for astronaut risk assessment. The results are expected to play a key role in helping to reduce the uncertainty associated with predicting the CNS risk for deep space exploration missions. This research is also designed to contribute to the identification of molecular targets for biological countermeasures to reduce CNS risks from space radiation.
Task Progress:	Our project continues to provide NASA with new mechanistic understanding of the risks for impaired neurocognitive performance and individual susceptibility for neurological diseases after exposure to HZE radiation. Our approach used an outbred rat model and: (a) employs state-of-the-art 'omics and tissue technologies to investigate CNS damage response functions in animals exposed to low-fluence HZE particles (i.e., <1 hit per cell), (b) links the type and magnitude of molecular defects with neurocognitive performance, and (c) validates the predicted damage responses in brain tissue. Our project focuses on radiation damage to the hippocampus (site of spatial and performance memories) and the choroid plexus (CP, small cauliflower-like tissue in the ventricles adjacent to the hippocampus that produces cerebral spinal fluid (CSF) and maintains blood-brain homeostasis). Our research distinguishes between two broad categories of risk: (1) mechanisms related to exposure dose, and (2) mechanisms of inter-individual variations in risks for neurocognitive performance, and neurological diseases (i.e., Alzheimer Disease (AD). Our working model is that neurocognitive damage after radiation involves direct cell damage to specific CNS target cells as well as indirect damage mediated by the irradiated microenvironment, such as abnormal CSF produced by an abnormal choroid plexus. The underlying hypothesis for this project is that exposure to space-relevant doses of HZE radiation (less than one ion per cell) induces abnormal molecular signaling and cellular damage in CP and hippocampus that determine risks and individual susceptibilities for neurocognitive deficits and neurological diseases such as AD. We confirmed that the CP is radiation sensitive with significant cellular and molecular damage after exposure to X-ray, 56Fe, or 12C radiation. We have discovered a time-course of molecular signaling changes in the CNS after exposure using transcriptomic, proteomic and metabolomics analyses of several brain regions (hippocampus, CP, CSF, and cortex) collected after whole body exposures, usually in individuals also tested on one or more behavioral assays. Persistent changes in gene and protein expression were associated with increased neuroinflammation, increased oxidative stress as well as certain neurological diseases and neurocognitive dysfunctions. Our experiments distinguish between radiation effects and normal aging. Cranial exposure of young adult rats to heavy particle ionizing radiation (HZE, 56Fe) causes neurocognitive deficits that persist to very old age, and the magnitude of the deficits are variable across animals. Beginning at 4 months and increasing in severity over 2 years after exposures, we observed histological changes of the CP including occluded endothelial vessels, increases in the frequency of amyloid deposits throughout the CNS, and decreased epithelial expression of proteins that normally prevent amyloid fibril formation in CNS. We also found increased frequencies of amyloid plaques in exposed animals but noted that the increase in plaque frequency was delayed 4-9 months after exposure. These findings suggest that higher doses of space radiation may increase the risks for Alzheimer-related pathologies in the brain, long after the exposure. Ongoing studies are in progress to understand the effects at space relevant doses of exposure. We also found that exposure to simulated space radiation diminished the ability of the CP to produce normal cerebral spinal fluid. Animals exposed to 10 and 100cGy 56Fe showed changes in many proteins associated with various neurological diseases, psychological disorders, neuronal cell death and survival. We also discovered molecular signaling

	changes that are associated with inter-individual variation in behavioral responses to HZE radiation, related to memory deficits and increased anxiety in rats. In summary, our research in a rat model identified persistent histological and molecular defects in the CNS tissues (CP, CSF, and hippocampus) that are different from normal aging and are associated with long-term neurocognitive dysfunction related to memory deficits, increased anxiety, as well as increased plaque formation long after HZE exposures.
Bibliography Type:	Description: (Last Updated: 06/19/2025)
Abstracts for Journals and Proceedings	Wyrobek AJ, Lowe XR, Rabin BM, Bhatnager S, Rademaker P, Northen T, Witkowska E, Straume T, Snijders A. "Molecular and cellular defects in the choroid plexus, CSF, and hippocampus of animals with persistent neurocognitive deficits after exposure to HZE radiation." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3321.pdf , Feb-2014
Abstracts for Journals and Proceedings	Wyrobek AJ, Rabin BM, Bhatnagar S. "Persistent neurocognitive decline, vascular and epithelial damage in the choroid plexus and beta amyloid plaques after cranial HZE radiation exposures." 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. , Sep-2014
Abstracts for Journals and Proceedings	Wyrobek AJ, Bhatnagar S, Rabin BM. "Persistent neurocognitive decline is ass ociated with vascular and epithelial damage to the choroid plexus and ß - amyloid plaques in an outbred rat model." American Society of Human Genetics 2014 Annual Meeting, San Diego, CA, October 18-22, 2014. American Society of Human Genetics 2014 Annual Meeting, San Diego, CA, October 18-22, 2014. p. 285. http://www.ashg.org/2014meeting/pdf/2014_ASHG_Meeting_Poster%20Abstracts.pdf ; accessed 2/23/15. , Oct-2014
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